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Advances in amino acid analysis of food proteins

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ABSTRACT

Accurate determination of amino acids in food proteins is essential for evaluating protein quality, nutritional value, and dietary adequacy. With increasing plant-based protein consumption and the essential role of amino acids in human health, there is a growing demand for accurate, high-throughput, and reproducible analytical techniques. However, many gold-standard techniques are time-consuming, labor-intensive, and expensive, limiting their suitability for high-throughput analysis. Consequently, the food protein industry urgently needs rapid, precise approaches for amino acid characterization. Developing such tools requires a clear and comprehensive understanding of both established techniques and the emerging methods shaping the field. This review comprehensively examines chromatographic, spectroscopic, and biosensor-based approaches for amino acid determination. Chromatography remains the benchmark for precise quantification due to high sensitivity and reproducibility, though it involves extensive sample preparation and longer analysis times. Spectroscopic methods enable rapid, often non-destructive, and high-throughput analysis, though matrix interferences and spectral overlap remain key challenges. Biosensors, integrating selective biorecognition elements with transducers, offer an emerging solution for portable, real-time, on-site monitoring in complex food matrices. By synthesizing current methodologies, analyzing their limitations, and highlighting research gaps, this review delineates a framework for developing efficient, sustainable, next-generation platforms for amino acid analysis in food proteins.

1. Introduction

Amino acids are organic compounds that form the building blocks of proteins in all living organisms. Physiologically, amino acids are essential for human health, serving both structural and regulatory roles. They constitute the primary substrates for protein synthesis, supporting the growth, repair, and maintenance of tissues and organs. In addition to their structural roles, amino acids act as precursors to neurotransmitters (e.g., serotonin from tryptophan) and hormones (e.g., thyroid hormones from tyrosine), thereby influencing numerous biochemical processes (Wu, 2009). Specific amino acids, such as leucine, are critical for stimulating muscle protein synthesis, whereas arginine supports ammonia

Abbreviations: IEC, Ion-Exchange Chromatography; HPLC, High-Performance Liquid Chromatography; GC, Gas Chromatography; UV-Vis, Ultraviolet-Visible; IR, Infrared; NMR, Nuclear Magnetic Resonance; MS, Mass Spectrometry; MIP, Molecularly Imprinted Polymer; LC-MS, Liquid Chromatography-Mass Spectrometry; OPA, O-Phthalaldehyde; Fmoc-Cl, 9-Fluorenylmethyloxycarbonyl Chloride; AQC, 6-Aminoquinolyl-N-Hydroxysuccinimidyl Carbamate; PITC, Phenylisothiocyanate; UHPLC, Ultra-High-Performance Liquid Chromatography; LC-MS/MS, Liquid Chromatography-Mass Spectrometry/Mass Spectrometry; GC-MS, Gas Chromatography-Mass Spectrometry; LC-HRMS, Liquid Chromatography-High-Resolution Mass Spectrometry; LC NMR, Liquid Chromatography-Nuclear Magnetic Resonance; LC-IR, Liquid Chromatography-Infrared Spectroscopy; LC-UV, Liquid Chromatography-Ultraviolet; UPLC-TQMS, Ultra-performance liquid chromatography tandem triple quadrupole mass spectrometry; SFS, Synchronous Fluorescence Spectroscopy; EEM, Excitation-Emission Matrices; MIR, Mid Infrared; NIR, Near Infrared; SERS, Surface-Enhanced Raman Spectroscopy; LDA, Linear Discriminant Analysis; PCA, Principal Component Analysis; TD ^{1}H NMR, Time-domain proton NMR; MALDI, Matrix-Assisted Laser Desorption/Ionization; ESI, Electrospray Ionization; PLSR, Partial Least Square Regression; XANES, X-ray Absorption Near-Edge Structure; EXAFS, Extended X-ray Absorption Fine Structure; EPR, Electron Paramagnetic Resonance; THz-TDS, Terahertz Time-Domain Spectroscopy; SPR, Surface Plasmon Resonance.

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detoxification (Layman, 2003; Wu, 2009). Deficiencies in essential amino acids compromise growth, immunity, and metabolic homeostasis, underscoring the importance of adequate dietary intake (Ling et al., 2023).

Proteins are made up of 20 different amino acids, nine of which are considered essential. The essential or indispensable amino acids such as histidine, isoleucine, leucine, lysine, methionine, phenylalanine, threonine, tryptophan, and valine cannot be synthesized by the human body, so they are obtained through diet. The remaining amino acids, though classified as non-essential or dispensable, are still necessary for protein synthesis and can be produced by the body through various biochemical processes (Barrett, 2000; Rutherfurd & Moughan, 2012).

Animal proteins, found in foods such as meat, poultry, fish, eggs, and dairy products, generally provide all essential amino acids in proportions suitable for human needs. This completeness makes them highly valuable in human nutrition. For example, whey protein from milk has a high concentration of leucine, an amino acid critical for muscle protein synthesis (Rieu et al., 2007). Similarly, egg protein is often considered the gold standard for protein quality because of its excellent amino acid profile and high digestibility (Layman & Rodriguez, 2009). In contrast, most plant proteins are deficient in one or more essential amino acids, making them incomplete proteins. For example, cereals are typically low in lysine, while legumes are low in sulfur amino acids such as cysteine and methionine. However, by combining different plant sources, such as rice and beans, it is possible to achieve a complete amino acid profile (Sá et al., 2020).

Amino acids are present in two different forms: free amino acids and protein-bound amino acids, which represent two distinct categories that differ in both functionality and analytical requirements. Free amino acids exist as unbound molecules in food matrices and play a direct role in taste, flavor development, and metabolic activity, whereas protein-bound amino acids are incorporated within peptide chains and contribute primarily to the nutritional and structural properties of proteins. The analysis of these two forms requires different approaches: free amino acids can often be measured directly with minimal sample preparation, while protein-bound amino acids typically require hydrolysis or enzymatic digestion to release individual amino acids prior to quantification. Distinguishing between these forms is therefore essential for selecting appropriate analytical techniques and for accurately interpreting compositional and functional data in food systems (EAKER, 1970; Synge, 1955; Wu et al., 2016).

The quantitative and qualitative determination of amino acids is central to assessing protein quality. Accurate amino acid profiling provides critical insights into amino acid balance, protein digestibility, and metabolic pathways relevant to human nutrition and food formulation (Rao & Poonia, 2023). As consumer demand for high-protein and functional foods increases, analytical techniques capable of precisely, rapidly, and cost-effectively detecting amino acids have become increasingly important in both research and industrial contexts. Moreover, with increased consumer interest in plant-based proteins, accurate characterization of their amino acid profiles, which is often incomplete and inferior to that of animal proteins, is highly desired by the health food sector.

The need for higher sensitivity and specificity in amino acid determination has necessitated the progression from classical biological and chemical assays to sophisticated instrumental methodologies. Early studies (1940–1960) by Snell, (1945) demonstrated that microbial growth responses are directly influenced by the availability of specific amino acids in defined media. This led to the development of microbial assays, where auxotrophic organisms were used to quantify amino acids based on growth response. Although, these methods were capable of quantitative amino acid analysis, its limitations include low specificity and matrix interference.

Chemical assays on the other hand, such as ninhydrin-based colorimetric methods, enabled faster detection by reacting with free amino groups, but were limited by poor selectivity and reproducibility (Sun et al., 2006). Subsequent developments focused on improving accuracy through hydrolysis-based release of amino acids from proteins, followed by chromatographic separation and quantification (Gehrke et al., 1985). The introduction of chromatographic techniques, particularly ion-exchange chromatography (IEC) and high-performance liquid chromatography (HPLC), marked a breakthrough, providing better resolution and quantitation of individual amino acids. These methods remain the benchmark for compositional analysis of complex food matrices. Gas chromatography (GC), although primarily applicable to volatile derivatives, has also made significant contributions to understanding amino acid-related flavour precursors and degradation pathways. Despite their high accuracy, chromatographic methods are often limited by the need for derivatization, lengthy analysis times, and the requirement for skilled operation, prompting the search for alternative, faster approaches (Kaspar et al., 2008).

To complement chromatography, spectroscopic techniques such as ultraviolet-visible (UV-Vis), fluorescence, vibrational, nuclear magnetic resonance (NMR), and mass spectrometry (MS) have been widely explored. These methods offer rapid, often non-destructive detection and can reveal both compositional and structural information. For example, UV-Vis and fluorescence spectroscopy are frequently used to monitor amino acid derivatives or intrinsic aromatic residues. In contrast, vibrational spectroscopy provides insights into protein conformational dynamics and secondary structure. NMR spectroscopy and MS-based techniques deliver unparalleled molecular-level resolution and sensitivity, facilitating both targeted and untargeted amino acid profiling. However, these methods require costly instrumentation, significant expertise, and may be affected by matrix interference in complex food systems. Nevertheless, their integration with chemometric and multivariate analysis has expanded their application to quality control, authentication, and process monitoring (Hassan et al., 2025; Kaspar et al., 2008; Kolev et al., 2008).

In recent years, biosensor technologies have emerged as a promising alternative for detecting amino acids, bridging the gap between laboratory-based analytical precision and field-level applicability. Biosensors integrate a biological recognition element (bioreceptors), such as an enzyme, aptamer, or molecularly imprinted polymer (MIP), with a transducer that converts biochemical interactions into measurable electrical or optical signals. Electrochemical biosensors, for instance, use amperometric or potentiometric principles to quantify amino acids based on redox reactions catalyzed by amino acid oxidases or dehydrogenases. Optical biosensors, employing fluorescence, surface plasmon resonance, or colorimetric detection, enable label-free, real-time monitoring. Their major advantages include portability, low reagent consumption, and the potential for in-situ or online food analysis. However, biosensors often face stability, reproducibility, and interference issues arising from complex food matrices (Dinu & Apetrei, 2022; Viswanathan et al., 2009; Yousefi et al., 2019).

The diversity of available analytical platforms underscores that no single technique can universally fulfill all analytical requirements. Chromatography provides the highest quantitative accuracy, spectroscopy offers speed and non-destructive analysis, and biosensors deliver real-time, user-friendly operation. The choice of technique depends on analytical objectives, sample type, and desired sensitivity. For instance, research applications may prioritize comprehensive amino acid profiling via liquid chromatography-mass spectrometry (LC-MS), whereas the industry may favour spectroscopic or biosensor-based methods for rapid quality assessment.

Despite the availability of chromatographic, spectroscopic, and biosensor-based techniques for amino acid analysis, there remains a gap in the comprehensive examination of the different aspects of amino acid detection and characterization provided by each method, as well as their practical applications. Many gold-standard techniques, although reliable, are often time-consuming, labor-intensive, or poorly suited to modern needs such as high-throughput, real-time, or on-site analysis. In parallel, emerging methods continue to develop, but their progress is limited by the absence of a consolidated framework that clearly outlines the principles, strengths, and limitations of existing technologies. Current literature frequently emphasizes individual techniques or specific applications, providing limited comparative insight into how different methods complement one another or differ in the type of analytical information they generate. This fragmented perspective limits informed selection of techniques for comprehensive food quality assessment. Accordingly, this review comprehensively analyzes chromatographic, spectroscopic, and biosensor-based techniques, highlighting how each technique contributes complementary data related to amino acid concentration, identity, structural features, dynamic behavior, and functional interactions. By doing so, it provides a roadmap to support future research and the development of next-generation approaches for amino acid detection. The following section examines chromatographic techniques, focusing on their principles, operation, and applications.

2. Chromatographic techniques for amino acid detection

2.1. Overview

Chromatographic techniques are the gold standard for determining amino acids in food proteins due to their high resolution, selectivity, and versatility. Generally, in food proteins, amino acids occur either in their free form or bound within peptides and proteins, which influence nutritional and sensory properties (Peace & Gilani, 2005). Precise quantification of these amino acids is pivotal in assessing protein quality, monitoring food authenticity, and guiding nutritional labelling standards. IEC was the first standardized method for amino acid analysis in foods, separating amino acids by charge using resin-based columns under controlled pH and ionic strength. Post-column ninhydrin derivatization enabled colorimetric or fluorometric detection, though the process was time-consuming and labour-intensive (Peace & Gilani, 2005; Yates et al., 2023).

The advent of HPLC revolutionized amino acid analysis by offering higher resolution, faster run times, and compatibility with various detectors (Fig. 1). HPLC introduced the use of pre- and post-column derivatization reagents, such as o-phthalaldehyde (OPA), 9-fluorenylmethyloxycarbonyl chloride (Fmoc-Cl), 6-aminoquinolyl-N-hydroxysuccinimidyl carbamate (AQC), and phenylisothiocyanate (PITC), to render non-chromophoric amino acids detectable through UV or fluorescence spectroscopy (Custodio-Mendoza et al., 2024; Lestari et al., 2022). Modern advances have led to the development of Ultra-High-Performance Liquid Chromatography (UHPLC) and Liquid Chromatography-Mass Spectrometry/Mass Spectrometry (LC-MS/MS) systems, offering ultra-fast separation, minimal solvent use, and enhanced sensitivity. These technologies enable the detection of multiple analytes and are increasingly applied in nutritional studies of foods (Fathana et al., 2021; Łozowicka et al., 2021; Mota et al., 2016; Weber, 2022). GC, although less favoured for polar compounds, remains indispensable for volatile amino acid derivatives in flavour research (Vilar et al., 2022).

2.1.1. Ion-Exchange Chromatography

Fig. 2 depicts a schematic diagram of a typical IEC. The process begins by preparing the ion-exchange column with a buffer and loading the sample, allowing charged molecules to bind to the resin. Unbound substances are washed away, and bound compounds are released by changing the salt concentration or pH. The separated fractions are then collected for further analysis. IEC is based on electrostatic attraction between charged analyte molecules and oppositely charged functional groups on the stationary phase of the resin. The strength of this ionic interaction depends on the molecule's charge, the pH of the mobile phase, and the ionic strength of the solution. The differential binding of molecules based on their net surface charge enables separation, as each component interacts with the resin to varying degrees. Post-column reaction with ninhydrin yields chromogenic derivatives, which are detected spectrophotometrically. Despite their exceptional accuracy and reproducibility, IEC systems require long run times (1–3 h per sample) and extensive manual handling, leading to their gradual replacement by faster LC-based approaches for high-throughput analysis (Cummins et al., 2017; Martin et al., 2013).

2.1.2. High-Performance Liquid Chromatography

HPLC is the most widely employed technique for amino acid analysis in food matrices, offering high sensitivity, reproducibility, and versatility. Fig. 3 depicts a schematic diagram of a general HPLC system. HPLC separates proteins based on their interactions with a stationary phase, typically a packed column and a mobile solvent. As the sample passes through the column under high pressure, proteins separate due to differences in polarity, charge, size, or binding affinity. Detection is achieved using UV, fluorescence, or mass spectrometry, with the integration of fluorescence detectors and ultra-sensitive photodiode array systems, enabling trace-level quantification and producing a chromatogram (Dai et al., 2014; Kambhampati et al., 2019; Mota et al., 2016). Because most amino acids lack strong native chromophores or fluorophores, derivatization is essential to form optically active components. Common reagents include OPA, which produces fluorescent derivatives detectable at 340/455 nm; Fmoc-Cl, which produces stable UV-absorbing or fluorescent products; and AQC or PITC to yield UV-detectable derivatives suitable for reversed-phase separation. Method performance depends on derivatization conditions, mobile phase composition, and gradient optimization. HPLC's speed, high throughput, and detector versatility have largely replaced classical IEC methods for routine amino acid analysis (de Souza et al., 2023; Gil et al., 2021; Kambhampati et al., 2019).

2.1.3. Gas Chromatography

GC, though traditionally used for volatile compounds, has specific applications in amino acid and peptide analysis after chemical derivatization. Fig. 4 shows a schematic diagram of a general GC. In this method, amino acids are converted into volatile, thermally stable derivatives via silylation, alkylation, or esterification, enabling separation in the gas phase. The vaporized sample is carried by an inert gas such as helium or nitrogen through a capillary column, where analytes are separated based on their differential interactions with the stationary phase. Compounds with weaker interactions elute earlier, while those with stronger interactions exhibit longer retention times, enabling effective resolution. Detection is typically performed using flame ionization detection or with a mass spectrometer (GC-MS – Gas Chromatography-Mass Spectrometry), the latter providing enhanced selectivity and structural information. GC techniques are useful for analyzing volatile amino acid derivatives in fermented and heat-treated foods, including enantiomers. However, complex derivatization and degradation of thermolabile amino acids limit their broader applicability (Peace & Gilani, 2005; Vilar et al., 2022).

2.2. Sample preparation and derivatization challenges

Sample preparation remains one of the most critical bottlenecks in amino acid chromatography, as food proteins are structurally complex and vary widely in solubility (Gunasekaran & Solar, 2012). Efficient hydrolysis is required to release constituent protein bound amino acids for accurate quantification. Conventional acid hydrolysis using 6 M hydrochloric acid at 110 °C for 24 h results in amino acid degradation and incomplete recovery, leading to errors in quantification (Lamp et al., 2018). Hydrolysis is often performed at multiple time intervals (e.g., 6–48 h) to improve accuracy. Shorter durations limit degradation but may not achieve complete release, whereas longer durations increase degradation. Labile residues such as tryptophan and cysteine are degraded while serine and threonine undergo partial loss (Zhang et al., 2026). In addition, hydrophobic amino acids including valine, leucine, and isoleucine are not fully released under these conditions due to slow

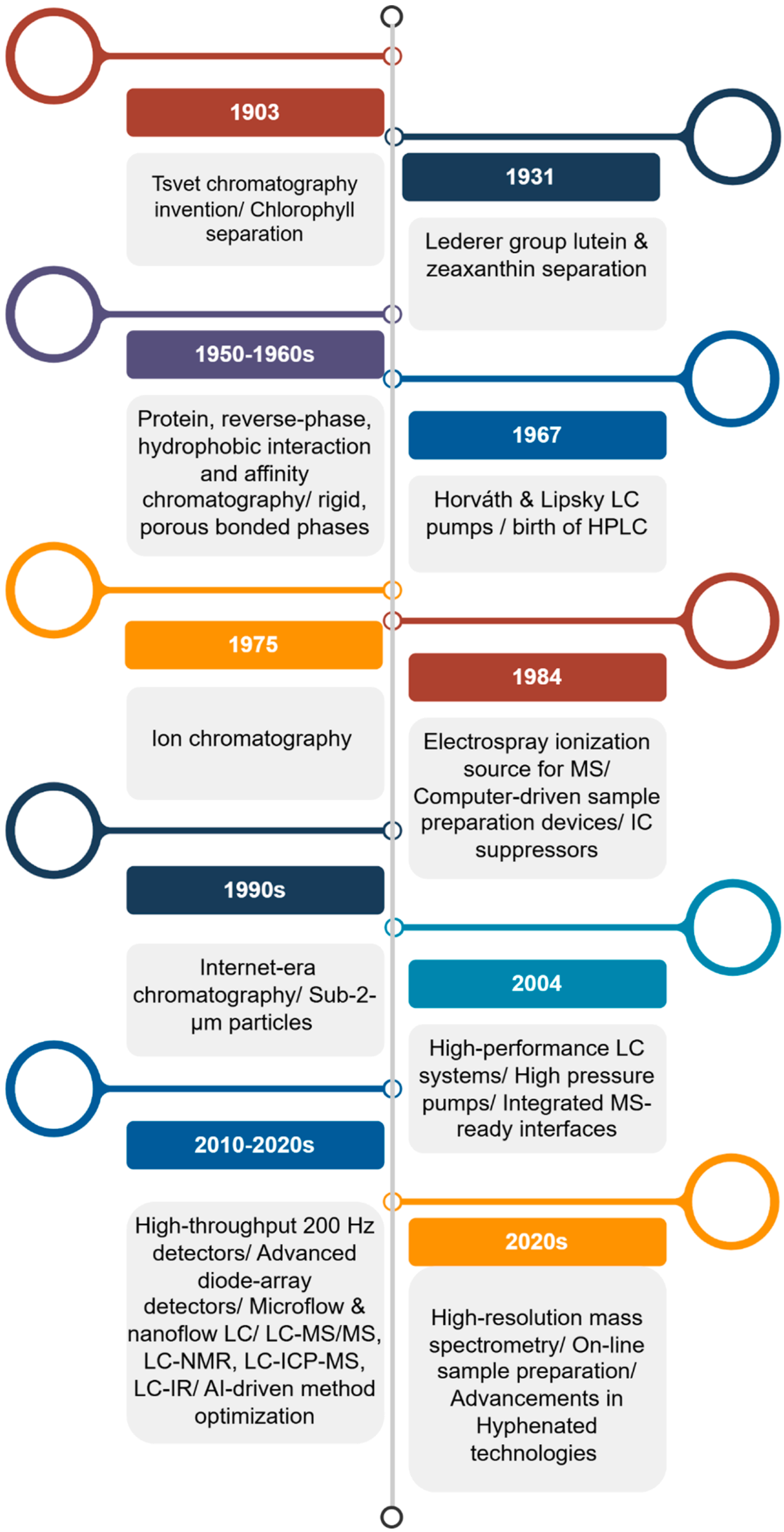

Fig. 1. Evolution of chromatographic systems (Adapted from (Baviskar et al., 2022; Câmara et al., 2022; More et al., 2025) LC - Liquid Chromatography, HPLC – High Performance Liquid Chromatography, IC – Ion Chromatography, MS – Mass Spectrometry, NMR – Nuclear Magnetic Resonance, ICP – Inductively Coupled Plasma, IR – Infrared.

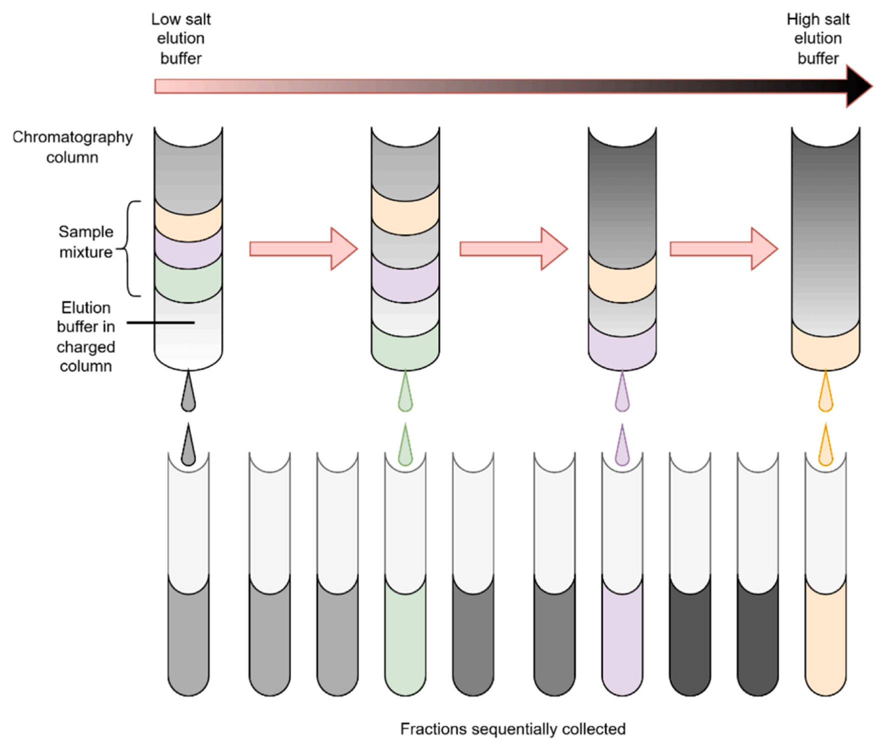

Fig. 2. Schematic diagram of Ion-Exchange Chromatography.

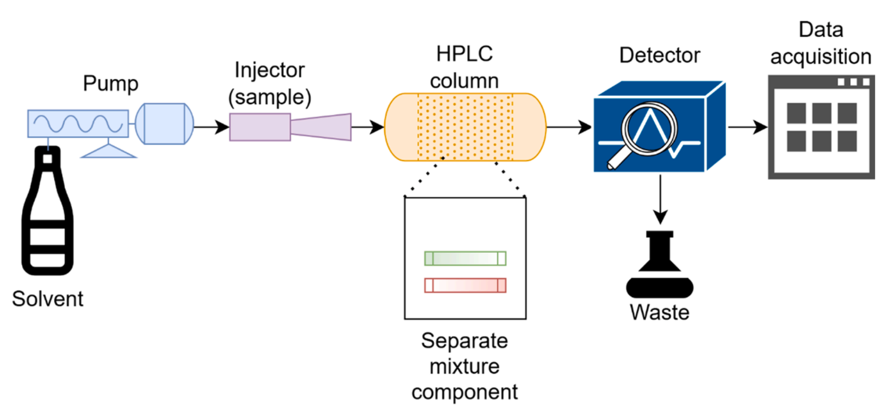

Fig. 3. Schematic diagram of High-Performance Liquid Chromatography.

peptide bond cleavage within structured regions of proteins, resulting in underestimation (Weber, 2022). Additional pre-treatment steps have been employed for accurate measurement of labile amino acids. Performic acid oxidation is used to stabilize sulfur-containing amino acids by converting methionine to methionine sulfone and cysteine to cysteic acid prior to hydrolysis (Yobi et al., 2023). Tryptophan, which is destroyed under acidic conditions, can determined separately using alkaline hydrolysis (Ansaf et al., 2023). A comparison study by Rehlund et al., (2025) found superior amino acid recoveries through enzymatic hydrolysis, aiding the quantification of asparagine, glutamine and improved recovery of tryptophan compared to alkaline hydrolysis. To overcome these drawbacks, modern methods, including enzyme and ultrasound-assisted hydrolysis and microwave-assisted digestion, have been developed to reduce reaction time and minimize amino acid decomposition (Custodio-Mendoza et al., 2024; Fountoulakis & Lahm, 1998).

Derivatization chemistry also plays a crucial role in analytical precision. It improves detection by increasing hydrophobicity and introducing chromophoric or fluorophoric groups, though it can induce signal drift or instability in specific amino acids. Pre-column derivatization offers higher throughput and reproducibility, while post-column techniques provide superior reaction consistency. The selection of derivatization reagents, such as OPA or PITC, depends on the amino acid profile, detection wavelength, and matrix characteristics (Yates et al., 2023). Moreover, matrix complexity in foods such as dairy and cereals necessitates the use of optimized extraction buffers and clean-up procedures, including solid-phase extraction, to minimize co-elution and enhance chromatographic resolution (Belarbi et al., 2021; Williams et al., 2023).

2.3. *Applications*

2.3.1. *Dairy and plant-based protein powders*

Chromatography provides robust tools for quality assessment of dairy and plant protein supplements. Johns (2021) developed a Liquid Chromatography – Ultraviolet (LC-UV) method for the quantification of methionine, cysteine, and tyrosine in milk- and plant-derived protein powders. The method involved acid hydrolysis (6 M hydrochloric acid,

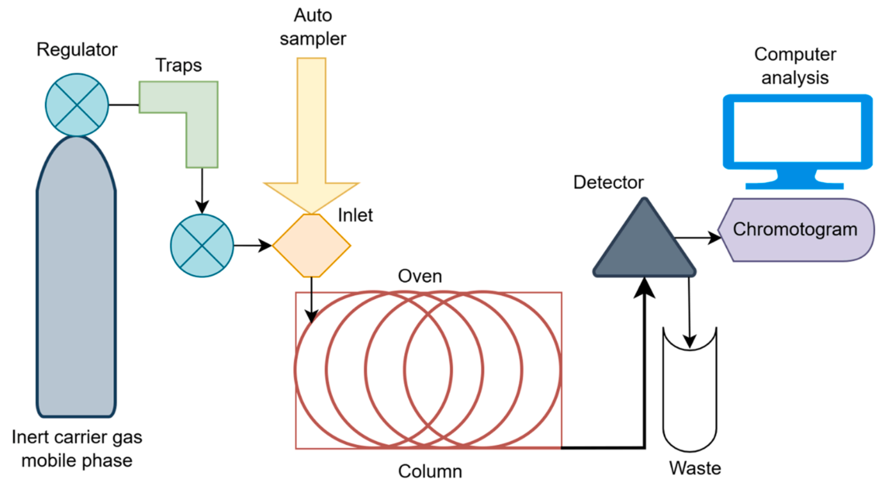

Fig. 4. Schematic diagram of a typical gas chromatography system.

110 °C, 24 h) in the presence of 3,3′-dithiopropionic acid and did not require derivatization. It demonstrated excellent linearity ($R^2 > 0.999$) and quantitation limits of 25 mg/100 g for methionine and tyrosine, and 50 mg/100 g for cystine, providing a robust and accessible approach for routine protein quality assessment. Hao et al., (2022) developed an authenticity-based method to detect adulteration in milk. They identified 46 peptides from trypsin-digested milk proteins, from which they selected marker peptides based on their digestibility and stability of milk proteins using an isotope-dilution ultra-performance liquid chromatography tandem triple quadrupole mass spectrometry (UPLC-TQMS). Using 22 authentic milk samples representing seasonal and regional variability, they established an authenticity threshold of 70.26 g of major milk proteins per 100 g of total protein. The method demonstrated high sensitivity, successfully detecting adulteration with as little as 10% added soy protein. Vigolo et al., (2022) studied the effect of factors affecting the milk protein composition using reversed-phase HPLC. The factors included three different preservatives (i.e., no preservative, hydrogen peroxide, Bronopol, and Azidiol), three freezing times at −20°C (0, 7, and 30 days), and two storage temperatures (5 and 21°C). The results indicated that prolonged freezing caused nonlinear protein degradation and storage temperatures altered key milk protein fractions (κ-casein, β-casein, α_{S1}-casein, β-lactoglobulin, and α-lactoalbumin). These chromatographic approaches are particularly useful in dairy authenticity, detecting amino acids in milk processing and storage conditions.

2.3.2. *Fruits, honey, and beverage matrices*

High-throughput LC methods have been tailored for complex liquid matrices such as fruits, wine, and honey. Kelly et al. (2010) reported a rapid LC method for the simultaneous determination of 24 amino acids and biogenic amines in fruits, wine, and honey, achieving a complete run (including column re-equilibration) in under 40 min without sample clean-up. The method was validated over a range of 0.25–10 mg/L, with repeatability and intermediate precision below 7% relative standard deviation, and accuracy between 97% and 101%. Limits of quantification ranged from 10 μg/L for compounds such as aspartic acid, glutamic acid, ethanolamine, and Gamma-Aminobutyric Acid, to 100 μg/L for tyrosine, phenylalanine, putrescine, and cadaverine. In a controlled LED-supplementation study conducted during morning and evening periods on 'MicroTom' tomato varieties, amino acid dynamics were investigated. Quantification was performed using LC-MS with reversed-phase chromatography and triple-quadrupole detection. 21 amino acids were extracted, chromatographically separated, and measured, while metabolic responses to red and blue LED treatments were monitored. Morning light supplementation markedly enhanced the accumulation of nearly all amino acids, whereas evening supplementation increased several essential and aromatic amino acids but showed weaker overall effects (Wang et al., 2022a, 2022b). Non-targeted GC-MS metabolomic profiling of five Indonesian mango cultivars identified 95 metabolites (47 annotated and 48 unknown metabolites). Principal Component Analysis (PCA) scores were used to associate the cultivars with taste and amino acids. For instance, cultivars with high impression scores in PCA (Arumanis 143, Gedong) accumulated sweetness-associated amino acids such as glycine, threonine, and tyrosine, whereas cultivars with low impression scores in PCA (Lalijiwo, Cengkir Indramayu) showed higher levels of sourness and bitterness-related amino acids, including aspartic acid, glutamic acid, and phenylalanine (Sato et al., 2021). Loh et al., (2021) studied the probiotic-fermented (*Lacticaseibacillus paracasei* Lpc-37 and *Saccharomyces cerevisiae* CNCM I-3856) unhopped beers. Quantification was performed by UPLC-MS/MS, and amino acid profiles were classified using hierarchical cluster analysis. Results indicated that *S. cerevisiae* unhopped beers had different amino acid profiles than *L. paracasei* Lpc-37 monoculture. Furthermore, they observed an increase in amino acid-derived metabolites, particularly branched-chain amino acids (leucine) and aromatic amino acids (phenylalanine, tyrosine, and tryptophan). Fermentation with *L. paracasei* Lpc-37 and *S. cerevisiae* CNCM I-3856, individually or synergistically, generated elevated levels of bioactive compounds.

2.3.3. *Plants and grains*

Chromatography elucidates protein fraction composition and amino acid profiles in cereals and pulses, informing breeding and quality assessment. Balindong et al., (2016) demonstrated that optimized protein extraction using 60% n-propanol and 5 M acetic acid, followed by HPLC analysis, effectively differentiated and quantified prolamin and glutelin in basmati (long and medium) and sushi rice varieties. These differences were associated with variations in eating quality, highlighting the relevance of precise amino acid profiling in cereal grain characterization. A streamlined extraction method combined with ninhydrin-based HPLC and capillary isotachophoresis enabled sensitive quantification of free amino acids for 12 gluten-free flours (corn, oat, soy, rice, pumpkin, millet, peanut, hemp seed, buckwheat, amaranth, pea, chickpea) compared with wheat flour. 14 amino acids including L-glutamic acid (highest in pea - 55.7 mg/100 g), L-methionine (up to 36.7 mg/100 g in pea), L-histamine (up to 28.3 mg/100 g in pea), L-valine (12.3 mg/100 g in peanut), L-lysine (3.76 mg/100 g in pea), L-tryptophan (8.36 mg/100 g in soy), L-phenylalanine (10.5 mg/100 g in peanut), L-tyrosine (24.5 mg/100 g in pumpkin), L-glycine (highest in hemp and pumpkin), and L-serine (notable in peanut) were detected

(Kowalska et al., 2021). Dahl-Lassen et al., (2018) employed UHPLC-MS with a single-quadrupole detector to analyze amino acids in various plant materials, such as red clover, ryegrass, lucerne, Arabidopsis, triticale seeds, spinach leaves, dried dog food pellets, and the National Institute of Standards and Technology (NIST)-1849a infant formula reference material. The method integrates classical protein hydrolysis and derivatization with rapid UHPLC separation and single-quadrupole detection. It achieves a 10-minute runtime, providing precision, accuracy, and sensitivity comparable to those of advanced, expensive mass spectrometers. Sensitivity was at least 10 times higher than that of fluorescence or UV-based detection. The technique accommodates sample sizes as small as 20 mg without losing reproducibility, enabling high-throughput, robust, and cost-effective amino acid profiling for both free and hydrolyzed proteins. Such chromatographic analyses guide cultivar selection, processing decisions, and nutritional labeling by providing reproducible, quantitative amino acid data across grain and legume matrices.

2.3.4. Meat and processed foods

Chromatography addresses both collagen markers and volatile flavor precursors in meat and processed products. Messia et al., (2008) combined microwave-assisted hydrolysis (20 min) with high-performance anion-exchange chromatography and pulsed amperometric detection to quantify 4-hydroxyproline, a collagen biomarker in the meat and meat-based products such as sausages. This method reduced hydrolysis from 24 h to 20 min while maintaining high precision (Relative Standard Deviation 0–6.4%) and without the use of pre- or post-column derivatization and non-toxic eluents. In another study, Corral et al., (2016) investigated the formation of odor-active sulphur and nitrogen compounds in dry fermented sausages. Their study linked the degradation of sulphur-containing amino acids (methionine and cysteine) and thiamine to the generation of volatile compounds, such as methionine to methional, thiamine to 2-acetyl-1-pyrroline, and cysteine to ethylpyrazine. Using solvent-assisted flavor evaporation coupled with GC-MS, nitrogen phosphorus detection, flame photometric detection, and olfactometry, they identified 17 volatile compounds associated with sensory attributes like cooked potato and nutty aromas. This work underscores the metabolic relevance of free amino acids generated by proteolysis during processing. These chromatographic strategies reveal biochemical pathways of proteolysis, oxidation and Maillard chemistry that shape texture and aroma during processing.

2.4. Advanced hyphenated and emerging techniques in chromatography

The integration of chromatography with spectrometric detection has enabled amino acid analysis with higher sensitivity and analytical throughput. Among these, LC coupled with tandem mass spectrometry and LC high-resolution MS (LC-HRMS) have become well-known, particularly in the areas of food chemistry and metabolomics (Weber, 2022; Zhao et al., 2021). These systems support simultaneous targeted and untargeted detection, providing mass accuracies below 5 ppm (Wang et al., 2021a, 2021b). LC-MS/MS use multiple reaction monitoring to improve quantification by filtering analytes based on unique fragmentation patterns (Li et al., 2023). This is crucial for distinguishing isobaric amino acids, modified derivatives, and degradation, especially in complex matrices such as fermented foods, protein hydrolysates, and thermally processed samples (Kim et al., 2025; Zhan et al., 2023).

Emerging chromatographic techniques extend these capabilities by enabling structural elucidation directly during the separation process. Liquid chromatography - nuclear magnetic resonance (LC NMR) and Liquid chromatography - infrared (LC-IR) represent advancements in direct molecular characterization (Baviskar et al., 2022). LC NMR, for instance, allows real-time acquisition of proton and carbon spectra from chromatographic eluents without prior isolation or crystallization (Kamal et al., 2021; Menard & Schug, 2025). This can be employed to identify unknown amino acid derivatives (Pérez-Victoria et al., 2022), Maillard reaction intermediates (Hammerl et al., 2021), and trace-level bioactive peptides in functional foods (Almoselhy et al., 2025). Similarly, LC-IR provides vibrational fingerprints that can differentiate amino acids with subtle structural differences, especially in sulphur amino acids, when analyzing stored or oxidized food samples. The variation in protein molecular weight distribution obtained by LC is used to measure protein oxidation. The rancidity or acidity of food is often measured using IR, which detects the secondary products of lipid oxidation (Geng et al., 2023).

The growing importance of sustainability has led to the development of "green chromatography." This focuses on reducing the environmental impact of analytical workflows through reduced solvent use, lab-on-a-chip columns, reduced mobile-phase toxicity, and the adoption of bio-derived solvents (Agrawal et al., 2022; Peyrin & Lipka, 2024). Micro-LC and nano-LC systems operate at flow rates of microliters per minute, drastically decreasing solvent consumption while maintaining chromatographic efficiency (Fedorenko & Bartkevics, 2023; Jesús et al., 2025). To minimize human error, these systems are supported by automated sample preparation techniques, such as online solid-phase extraction, microextraction using packed sorbents, and digital solvent delivery pumps (Gallardo et al., 2025).

Integration with digital workflows has broadened the utility of advanced chromatographic systems. Modern LC platforms now incorporate machine learning algorithms for peak recognition, baseline correction, and spectral deconvolution. These computational additions are valuable for large-scale food authenticity testing, where rapid detection of adulteration or markers of protein degradation is needed. Additionally, cloud-connected chromatographic instruments facilitate remote monitoring, real-time quality assurance, and centralized data archiving (Ali et al., 2025; Balamurugan et al., 2025; Das and Kenneth M. Merz, 2025).

Several limitations are posed by advanced hyphenated chromatographic systems, including the costs of maintenance, operational training, and standardization. Matrix effects remain a persistent problem in LC-MS workflows, with ion suppression, adduct formation, or co-eluting carbohydrates and lipids (Eddahby et al., 2025). Furthermore, micro-LC and nano-LC systems often suffer from column clogging, back-pressure instability, and reduced robustness when handling viscous or particulate-rich matrices (Amir Hamzah et al., 2025). These challenges highlight the need for continued development in areas such as standardized workflows, improved microfluidic durability, and greater integration of machine-learning-based spectral correction and peak deconvolution tools.

2.5. Summary of chromatographic techniques

Chromatographic techniques remain the cornerstone of amino acid analysis in food systems due to their exceptional precision, reproducibility, and versatility across complex matrices. Methods such as IEC, HPLC, and GC have evolved to deliver faster separations, enhanced automation, and improved sensitivity through the use of advanced detectors and derivatization strategies. Table 1 summarizes the various chromatographic techniques. Recent innovations emphasize greener analytical practices, including reduced solvent use and miniaturized column systems, aligning with sustainability goals. However, chromatographic methods often require extensive sample preparation, derivatization, and longer run times, which may limit their applicability in high-throughput or on-site food monitoring. To address these challenges, spectroscopic techniques have emerged as efficient, cost-effective, and complementary methods for analyzing amino acids in complex food matrices. In the next section, principles, working mechanisms, and applications of various spectroscopic techniques are discussed.

Table 1
Summary of chromatographic techniques.

Technique	Sample preparation complexity	Sample Preparation	Sensitivity (LOD, μg/mL)	Throughput	Preferred matrix	Relative cost
IEC	High	Acid hydrolysis (6 M HCl, 6–24 h), neutralization, filtration, buffer dilution	1–5	Low	All hydrolyzed proteins	Moderate
HPLC (UV/FLD)	Medium	Acid hydrolysis ± derivatization (OPA/ Fmoc-Cl), filtration (0.22 μm), dilution, internal standard	0.1–1.0	Moderate	All hydrolyzed proteins	Moderate
GC-MS	High	Acid hydrolysis, derivatization (silylation, alkylation, or esterification), drying, solvent extraction	0.5–2	Moderate	Fermented or volatile amino acids	High
LC-HRMS / LC NMR/LC-IR	Variable	Minimal preparation or hydrolysis (matrix dependent), filtration / centrifugation, dilution, internal standard	0.01–0.1	Moderate-High	Specialized foods	Very High

LOD - limit of detection, IEC - Ion-Exchange Chromatography, HPLC - High-Performance Liquid Chromatography, UV - Ultraviolet, FLD - Fluorescence detectors, GC-MS - Gas Chromatography-Mass Spectrometry, LC-HRMS - Liquid Chromatography High-Resolution Mass Spectrometry, LC NMR - Liquid Chromatography-Nuclear Magnetic Resonance, LC-IR - Liquid Chromatography – Infrared, OPA – O-Phthalaldehyde, Fmoc-Cl – 9-Fluorenylmethyloxycarbonyl Chloride

3. Spectroscopic techniques

3.1. Overview of spectroscopic detection

Spectroscopic techniques have gained increasing attention in amino acid and food protein analysis due to their ability to provide rapid, non-destructive, and highly informative molecular insights with minimal sample preparation. Although chromatographic methods remain the benchmark for precise quantification, spectroscopic methods offer complementary advantages by enabling in situ, real-time analysis. They provide valuable information on molecular structures, conformational changes, and physicochemical interactions, supporting the evaluation of multi-component systems and quality assurance in food authenticity and processing studies.

Spectroscopy comprises analytical techniques that study the interaction between electromagnetic radiation and matter to reveal molecular composition and structure. A spectrometer measures how molecules absorb, emit, or scatter light at characteristic wavelengths, producing a spectrum that reflects their chemical and physical properties. A typical spectroscopic setup consists of a light source, a wavelength selector (e.g., a monochromator), a sample holder, and a detector. UV-spectroscopy is one of the earliest spectroscopic techniques used in food analysis. Fig. 5 shows a schematic diagram of a UV-spectrometer. Initially, the sample is illuminated by a light source to ensure constant radiation power across the wavelength range. The light then passes through the monochromator, which selects a narrow band of wavelengths from the broader spectrum. Prisms or diffraction gratings are commonly used as monochromators. Next, the monochromatic light passes through the sample placed in the sample holder. Sample holders are typically made of quartz or fused silica cuvettes, as these materials allow transmission of UV and visible light. Finally, the transmitted light reaches the detector, where it is converted into an electrical signal and measured as absorbance versus intensity. These spectral patterns provide both qualitative and quantitative insights into molecular features such as functional groups and concentration, enabling accurate identification and analysis of amino acids and other biomolecules in food systems (Karoui & Blecker, 2010; Mandru et al., 2023).

In food proteins, spectroscopic techniques are widely used to evaluate protein content, amino acid composition, and protein functionality in various matrices. These methods include UV-Vis spectroscopy, which measures electronic transitions in chromophores to quantify proteins and specific amino acids (Mandru et al., 2023); fluorescence spectroscopy, which uses intrinsic or extrinsic fluorophores to study protein interactions, conformational changes, and thermal stability (Karoui, 2018); infrared (IR) spectroscopy, including mid-IR (MIR) and near-IR (NIR), which probe vibrational transitions to assess secondary structure, functional groups, and compositional attributes (Wang & Paliwal, 2007); Raman and surface-enhanced Raman spectroscopy (SERS), which provide vibrational fingerprints for structural characterization and trace-level detection (Findlay et al., 2025); and nuclear magnetic resonance (NMR) spectroscopy, which delivers detailed information on molecular dynamics, hydration, and structural organization (Parlak & Guzeler, 2016). Collectively, these techniques enable rapid, non-destructive, multicomponent analysis. When combined with chemometric or multivariate tools, they can provide both quantitative and qualitative insights that are critical for assessing food quality, authenticity, and protein functionality assessment. The following section provides an overview of the spectroscopic techniques used to analyze food proteins.

3.2. Optical spectroscopy

3.2.1. UV-Vis spectroscopy

UV-Vis spectroscopy is a widely applied technique for determining

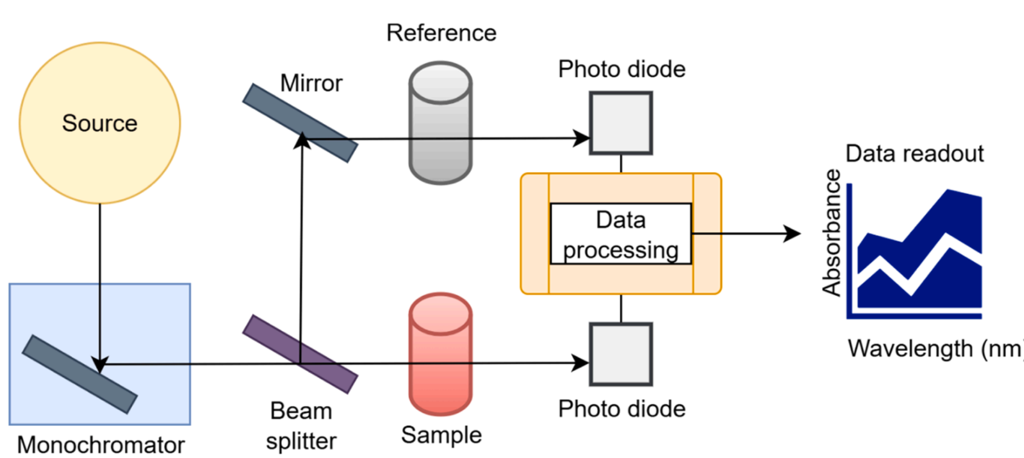

Fig. 5. Schematic diagram of a UV spectrometer.

total protein content as well as for quantifying specific amino acids in the wavelength range of 200–800 nm. Its principle is based on the electronic transitions of aromatic residues, primarily tryptophan, tyrosine, and phenylalanine, which absorb strongly at approximately 280 nm. Conventional protein assays, such as the Biuret, Bradford, and Lowry methods, measure changes in absorbance resulting from chemical reactions involving peptide bonds or amino groups. For individual amino acids, derivatization with reagents such as ninhydrin or AQC enables direct optical detection and quantitative analysis. While UV-Vis spectroscopy is rapid, simple, and cost-effective, its precision can be compromised by matrix turbidity, overlapping chromophores, and oxidation, particularly for low-abundance amino acids. Moreover, its applicability is inherently limited to a small subset of amino acids, predominantly aromatic residues. Therefore, UV-Vis spectroscopy alone may not provide comprehensive amino acid profiling, and this limits its ability to replace chromatographic methods. Coupling UV-Vis detection with chromatographic separation techniques, such as HPLC and UHPLC, enhances specificity and sensitivity, enabling more reliable quantification. This combination provides an efficient balance between throughput and analytical accuracy, making it a versatile tool in food protein and amino acid analysis (Mandru et al., 2023; Schmid, 2001).

3.2.2. Fluorescence spectroscopy

Fluorescence spectroscopy offers high sensitivity and selectivity, particularly for aromatic amino acids like tryptophan and tyrosine. Intrinsic fluorescence allows detection of subtle conformational changes and alterations in the local microenvironment of proteins. The use of extrinsic probes, including dansyl chloride and FITC, extends the dynamic range and facilitates analysis in complex food matrices (Lakowicz, 2006). Fluorescence spectroscopy is especially valuable for monitoring oxidative modifications, protein denaturation, and aggregation in dairy and meat products. Its high signal-to-noise ratio and the ability to perform front-face measurements make it suitable for opaque or turbid samples, enabling rapid, non-destructive analysis (Hougaard et al., 2013; Karoui, 2018). However, similar to UV–Vis spectroscopy, fluorescence techniques are restricted to a limited number of naturally fluorescent amino acids and often require labeling or derivatization for broader applicability, which can increase complexity and reduce throughput. These capabilities make fluorescence an indispensable tool for quality control, process monitoring, and functional characterization of proteins in diverse food applications.

3.2.3. Vibrational spectroscopy

Vibrational spectroscopy, including NIR, MIR and Raman spectroscopy, characterizes the molecular vibrations of amino acids and proteins. Absorbance features in the protein region (1500–1700 cm^{-1}), specifically the amide I (around 1650 cm^{-1}) and amide II (around 1550 cm^{-1}) bands, provide valuable information on protein secondary structure and its relationship to amino acid composition (Carbonaro & Nucara, 2010). By applying chemometric techniques such as multivariate analysis, these spectra can be converted into predictive models for functional attributes, protein and amino acid content, thereby reducing dependence on traditional wet-chemistry methods (Sahar et al., 2019). NIR spectroscopy is best suited for rapid, non-destructive analysis of bulk food samples that require moisture, protein, and overall compositional estimation, making it ideal for online or in-situ quality monitoring. Nevertheless, NIR is associated with relatively low spatial resolution and limited penetration depth, which can restrict its effectiveness for heterogeneous samples. On the other hand, MIR spectroscopy is well-suited when higher molecular specificity is required, as its fundamental vibrational bands provide more accurate identification of functional groups and detailed chemical profiling. In practice, NIR offers speed and robustness for routine screening, while MIR delivers superior analytical precision for targeted amino acid, protein, or structural characterization. When combined with regression or classification models or other machine learning algorithms, NIR and MIR enable accurate prediction of amino acid and protein profiles in milk, cereals, pulses, and composite food products, supporting high-throughput quality evaluation in food processing (ElMasry et al., 2012; Mundhada et al., 2022; Sivakumar et al., 2022, 2025). However, these approaches rely heavily on robust calibration models that are highly specific to individual food matrices, and without appropriate calibration datasets, accurate analysis is not feasible. In addition, strong matrix effects, including overlapping spectral features, light scattering from turbid samples, and interference from pigments or other metabolites, can complicate quantitative interpretation and require extensive preprocessing.

Raman spectroscopy is a rapid, non-destructive vibrational technique that detects unique molecular vibrations generated when laser light interacts with a sample. The resulting scattered light produces characteristic spectral peaks corresponding to specific chemical bonds, enabling direct, label-free analysis of proteins, lipids, and carbohydrates. Protein-related vibrational bands in the Raman spectrum reflect molecular structure, allowing assessment of secondary structures and conformational changes (Rygula et al., 2013). It is worth noting that NIR spectroscopy relies on changes in dipole moment, whereas Raman spectroscopy relies on changes in molecular polarizability; thus, the two techniques provide complementary, non-redundant information. Additionally, Raman spectroscopy is well-suited for analyzing aqueous solutions (Cozzolino, 2022; Li-Chan et al., 2002).

A major limitation of Raman spectroscopy is its inherently weak signal, with only about one photon in a million contributing to the Raman response. To address this, enhancement strategies such as SERS, stimulated Raman scattering, coherent anti-Stokes Raman scattering, and resonance Raman spectroscopy have been developed, enabling signal amplification by several orders of magnitude and allowing trace-level detection of amino acids and proteins (Cozzolino, 2022; Zong et al., 2018). Despite these advances, issues such as fluorescence interference, sample heating, and variability in enhancement substrates (e.g., in SERS) can affect reproducibility and limit routine application. Recent studies have further integrated Raman and SERS with multivariate statistical approaches, including PCA and linear discriminant analysis (LDA), to identify food adulteration, determine protein content, and monitor thermal or oxidative changes (Liu et al., 2020a, 2020b; Wang et al., 2021a, 2021b). This combination of molecular spectroscopy and advanced data analytics represents a modern, powerful approach in food analysis, transforming high-dimensional spectral data into meaningful biochemical insights that support authenticity verification and quality assurance.

3.2.4. Nuclear magnetic resonance

NMR spectroscopy provides molecular-level insights into protein composition, structure, and interactions by measuring the relaxation behavior of atomic nuclei in a magnetic field. It enables the study of protein hydration, conformational dynamics, and molecular mobility in both liquid and solid food systems. Time-domain proton NMR (TD ^{1}H NMR) is beneficial for monitoring water distribution and phase transitions during food processing, such as baking or meat gel formation (Pycarelle et al., 2020; Riley et al., 2022; Zheng et al., 2021). High-resolution NMR enables the quantification of individual amino acids and the identification of chemical modifications (Ma et al., 2025). These measurements are non-destructive and provide real-time information that complements conventional chromatographic and thermal techniques. However, NMR instrumentation is expensive, requires specialized expertise, and often has lower sensitivity compared to mass spectrometry, which can limit its applicability for trace-level analysis in complex food matrices. When integrated with rheological and microscopic analyses, NMR helps establish relationships between molecular organization and macroscopic texture, offering a predictive framework to optimize protein functionality, improve formulation stability, and enhance the sensory and structural qualities of food products (Chen et al., 2021; De Kort et al., 2017).

3.3. Mass spectroscopy

MS-based methods, although generally destructive, rely on ionization and mass-to-charge (m/z) detection. Among the primary ionization techniques, Matrix-Assisted Laser Desorption/Ionization (MALDI) and Electrospray Ionization (ESI) have enabled highly precise profiling of amino acids and peptides from food protein hydrolysates (Mamone et al., 2009; Terada et al., 2024). MALDI-MS produces characteristic spectral barcodes of peptides or free amino acids, which can be used to identify and authenticate protein sources in complex food systems. In contrast, ESI-MS, particularly when coupled with UHPLC-ESI-MS/MS, allows simultaneous quantification of multiple amino acids with exceptional sensitivity and selectivity (Bùi et al., 2023). For instance, recent UHPLC-triple quadrupole MS workflows have quantified up to 18 amino acids in meat matrices for nutritional assessment (Wang et al., 2024). MS-based analytical techniques offer high precision and depth of analysis but require sophisticated instrumentation and extensive sample preparation. Additionally, matrix effects, ion suppression, and the need for careful method optimization can affect quantification accuracy, particularly in complex food systems.

3.4. Chemometric and multivariate data analysis

Spectroscopic measurements produce high-dimensional, multicollinear data that require advanced computational tools for interpretation. Chemometric modeling techniques, including PCA, partial least squares regression (PLSR), discriminant analysis, and emerging machine learning algorithms, are now essential for extracting relevant information from amino acid-related spectra (Hassan et al., 2025; Kang et al., 2022; Leite et al., 2025). These approaches enable pattern recognition, outlier detection, and quantitative calibration without the need for physical separation. For example, PCA can classify protein-rich foods based on amino acid fingerprints (Sivakumar et al., 2023, 2024b), whereas regression models directly predict nutritional indices from MIR or NIR spectra (Mundhada et al., 2022; Sivakumar et al., 2023). The integration of machine learning methods, such as support vector machines (SVMs) and convolutional neural networks (CNNs), enhances predictive performance on bulk samples, including wheat grains (Kılıçarslan & Kılıçarslan, 2024; Lingwal et al., 2021; Yasar, 2024; Gorgannejad et al., 2026). Beyond quantification, chemometrics enables authenticity testing, traceability, and prediction of functionality (Rodionova et al., 2024). However, these approaches require large, well-annotated datasets that capture variability across food matrices, and such datasets are often costly and difficult to obtain. Also, calibration transfer between instruments remains a significant limitation, prompting interest in hybrid models and cloud-based frameworks for universal spectral analysis.

3.5. Sample preparation

Sample preparation for spectroscopic analysis in food systems varies depending on the technique, the food matrix, and the desired information. Some techniques, such as vibrational spectroscopy, can often be applied directly to solid or liquid food samples with minimal preparation, enabling rapid, non-destructive, and in situ measurements for free amino acids. For instance, whole grains, flours, or homogenized dairy products can be analyzed without prior extraction (Su & Sun, 2018; Wu & Sun, 2013), whereas SERS may require deposition of the sample onto a nanostructured substrate to enhance signal detection (Findlay et al., 2025). In contrast, techniques such as UV-Vis spectroscopy and fluorescence spectroscopy typically require the sample to be dissolved or suspended in an appropriate solvent or buffer to ensure homogeneity and accurate absorbance or emission measurements to measure free amino acids (Lakowicz, 2006; Mandru et al., 2023). Similarly, LC-UV or LC-MS-based methods often require protein extraction, hydrolysis, derivatization, or filtration to isolate the target protein bound amino acids or peptides from complex food matrices (Lucci et al., 2017). NMR spectroscopy, while versatile, may also require sample dissolution in deuterated solvents, pH adjustment, or homogenization for liquid-state measurements (Cao et al., 2021). The choice of sample preparation method must balance the analysis goals, such as speed, sensitivity, or structural fidelity, with the complexity and composition of the food matrix. Inadequate sample preparation can impact spectral quality, reproducibility, and quantitative accuracy.

3.6. Applications

3.6.1. Cereals and pulses

UV-Vis spectroscopy has been widely used to analyze the protein and amino acid composition of pulses such as black-eyed bean, Bambara groundnut, soybean, and African black bean, as well as in oilseeds such as Dika nut and melon seed (Okoronkwo et al., 2017). NIR has shown strong predictive capability for protein and amino acid profiling in lentil seeds (Hang et al., 2022), brown rice flour (Xie et al., 2023), and whole soybeans (Kovalenko et al., 2006), enabling rapid compositional screening in cereals and pulses. MIR spectroscopy, combined with machine learning, has also been employed to analyze blends of pulses, cereal grains, and pseudocereals, accurately predicting protein content and distinguishing between gluten-free and non-gluten-free samples (dos Santos et al., 2025; Sivakumar et al., 2023, 2024a, 2025). Additionally, Attenuated Total Reflectance-MIR spectroscopy has been used to characterize the secondary structures of wheat proteins, revealing the predominance of α-helices and β-turns in albumins, globulins, and glutenins (Nadeak et al., 2025).

3.6.2. Dairy

Fluorescence spectroscopy has proven highly effective for monitoring heat-induced changes in milk. Front-face fluorescence was applied to predict lactulose formation in heat-treated milk (Ayala et al., 2017) and to classify pasteurized milk according to processing intensity (Hougaard et al., 2013). It has also been used to investigate interactions between anthocyanins such as pelargonidin and dairy proteins, showing strong static quenching and thermodynamically driven binding affinities (Arroyo-Maya et al., 2016). LC-UV spectroscopy has facilitated accurate quantification of sulphur amino acids in milk-derived protein powders (Johns, 2021), while MIR spectroscopy has recently been used to predict amino acid composition in bovine milk, demonstrating moderate predictive accuracy for key amino acids such as glutamic acid, leucine, and methionine (Chu et al., 2025).

3.6.3. Meat

Low-field NMR has been used to study hydration dynamics in meat batters, identifying distinct relaxation components corresponding to bound and free water, as well as fat mobility (Shao et al., 2016). MS has been instrumental in detecting species adulteration in meat products (Zhong et al., 2022). Raman spectroscopy has been applied to monitor chicken meat spoilage by tracking spectral changes associated with amino acid degradation and protein denaturation (Zając et al., 2017).

3.6.4. Bakery Products

Time-domain proton NMR (TD ^{1}H NMR) has emerged as a valuable tool in studying protein and starch transformations in bakery systems. It has been used to monitor molecular dynamics in sponge cake batters and bread matrices, revealing how proteins and starch transition from rigid to mobile states during baking. Cakes containing exogenous lipids exhibited increased protein polymerization and improved structure, while bread studies demonstrated that water absorption, amylose leaching, and amylopectin melting significantly influence crumb setting and texture (Nivelle et al., 2019; Pycarelle et al., 2020). Additionally, NMR studies on soy protein isolate and wheat gluten blends have linked water distribution to viscoelastic behaviour, offering insights into dough elasticity and texture optimization (Dekkers et al., 2016).

3.6.5. Oilseeds

UV-Vis spectroscopy has been applied to determine protein and amino acid profiles in oilseeds such as Dika nut, groundnut, melon seed, and castor oil seed (Okoronkwo et al., 2017). Capillary electrophoresis coupled with mass spectrometry has been used to identify non-protein amino acids in vegetable oils, successfully detecting adulteration of olive oil with as little as 2% soybean oil (Sánchez-Hernández et al., 2011). Raman spectroscopy has also been employed to study structural changes in semi-refined flaxseed protein extracts subjected to pulsed electric field-assisted extraction, revealing alterations in α-helical and β-sheet structures (Punzalan et al., 2025).

3.7. Emerging trends and knowledge gaps

Advanced spectroscopic techniques, such as X-ray Absorption Near-Edge Structure (XANES), Extended X-ray Absorption Fine Structure (EXAFS), Electron Paramagnetic Resonance (EPR), and Terahertz Time-Domain Spectroscopy (THz-TDS), have become promising tools for probing molecular structure, functional groups, and dynamic interactions within complex food matrices. Their high sensitivity, non-destructive nature, and element- or state-specific resolution make them particularly valuable for accurately characterizing amino acids and protein-derived components in modern food analysis. XANES and EXAFS offer detailed insight into the local chemical environment of specific elements by tracking energy-dependent X-ray absorption patterns. These techniques elucidate oxidation states, coordination geometries, metal-protein interactions, and subtle structural modifications that influence amino acid behaviour, protein conformation, and nutrient bioavailability in foods (Ceko et al., 2014; Clemente et al., 2021; Prange et al., 2001; Schiell et al., 2025; Wojtaszek et al., 2025). Such information is crucial for understanding the effects of processing, storage, and fortification on amino acid stability. EPR spectroscopy provides a complementary perspective by selectively detecting unpaired electrons in free radicals and paramagnetic centers. This makes EPR ideal for monitoring oxidative pathways involving amino acids during thermal processing, lipid oxidation, enzymatic browning, or storage. Its ability to capture transient radical species, particularly when enhanced by spin-trapping or spin-labelling methodologies, allows researchers to track structural transitions and degradation routes of amino acids such as cysteine, proline, and alanine (Akram et al., 2016; Barba et al., 2020; Escudero et al., 2019; Jiang et al., 2018; Karakirova, 2023; H. Li et al., 2021; Zhang et al., 2015). This is especially relevant for protein-rich or oxidative-stress-prone foods where amino acid integrity is a marker of freshness and quality.

THz-TDS offers additional analytical power by probing low-frequency vibrational modes (0.1–10 THz) that reflect hydrogen-bond networks, crystalline arrangements, and weak intermolecular interactions. Many amino acids and peptides exhibit characteristic THz absorption signatures, enabling their discrimination and the detection of structural changes caused by hydration, heat processing, or adulteration (Afsah-Hejri et al., 2019; Y. Liu et al., 2019; Lu et al., 2016; Zhang et al., 2025; Zhang et al., 2017). These spectral fingerprints make THz-TDS a promising tool for rapid, label-free assessment of amino acid composition in protein-based ingredients and complex food systems.

Despite such advances, several critical gaps remain before these spectroscopic methods can achieve wider routine adoption in food laboratories. Strong matrix effects, including overlapping spectra, scattering from turbid samples, pigments, and interactions with other metabolites, still complicate quantitative interpretation and often require extensive preprocessing. Many amino acids lack prominent native spectroscopic signatures, necessitating derivatization or indirect detection strategies that reduce throughput. Portable instruments currently lag behind benchtop systems in sensitivity, stability, and reproducibility, while cross-instrument variability limits the transferability of calibration models. The absence of standardized spectral libraries, validated reference materials, and universal protocols impedes comparability across studies and slows regulatory acceptance. Additionally, machine-learning-driven spectroscopy requires large, well-annotated datasets that capture diverse food matrices, but such datasets remain expensive and difficult to curate.

3.8. Summary of spectroscopic techniques

Table 2 compares various spectroscopic techniques. UV-Vis and fluorescence spectroscopy remain the most widely applied for routine amino acid detection due to their simplicity, affordability, and compatibility with derivatized analytes. Vibrational spectroscopy provides valuable structural insights and enables rapid, non-destructive monitoring of protein-related changes during food processing. NMR spectroscopy offers unparalleled structural resolution and quantitative capability but is limited by high operational cost and long analysis times, making it more suitable for research applications. MS delivers exceptional sensitivity and specificity, allowing precise identification and quantification of amino acids even within complex food matrices. However, their extensive sample preparation and instrumentation requirements restrict widespread industrial use.

Despite their analytical power, spectroscopic methods often require advanced data processing and calibration models, which limit their portability and real-time application. To overcome such restrictions, biosensor technologies have emerged that merge the biochemical specificity of enzymatic or affinity-based recognition with spectroscopic transduction. This convergence enables the development of a new generation of miniaturized, cost-efficient, and user-friendly platforms for detecting amino acids directly in complex food matrices. The following section provides an overview of the principles and applications of biosensor technologies.

4. Biosensor-based techniques

4.1. Overview of biosensing for amino acid detection

Biosensors are analytical devices that integrate a biological recognition element (bioreceptor) with a transducer and a signal-processing component to convert a biochemical interaction into a measurable output (e.g., electrical, optical, or mechanical) as shown in Fig. 6. A biosensor detects a target analyte by its interaction with a bioreceptor, such as an enzyme, aptamer, or molecularly imprinted polymer *(as explained in detail in Section 4.2)*, thereby providing high specificity. This interaction is converted into a measurable signal by a transducer, either electrical (based on changes in electric charge or electric impedance), electrochemical (based on changes in current, voltage, or impedance with a reference electrode) or optical (based on shifts in absorbance, fluorescence, or refractive index) *(as explained in detail in Section 4.3)*. The resulting signal is amplified and interpreted by an onboard processor using calibration or embedded algorithms. Finally, the processed data are displayed on a digital interface or on a smartphone, enabling rapid, user-friendly detection. In the context of amino acid detection in food proteins, biosensors offer several key features, including high selectivity (via the bioreceptor), rapid response, potential for miniaturization, and portability. Also, they are well-suited for in-situ or point-of-use monitoring, which is increasingly required in protein quality and food authenticity measurements (Malik et al., 2023a).

Unlike traditional chromatographic or spectroscopic methods, which provide excellent accuracy but require lengthy sample preparation, derivatization, expensive instrumentation, and centralized laboratory facilities (*as detailed in Sections 2 and 3*), biosensors offer a more streamlined alternative. They enable rapid screening, real-time monitoring, and on-site detection of specific amino acids or their derivatives in protein hydrolysates or food extracts. For example, aflatoxin, a polyketide compound produced by fungal species during the secondary metabolic pathway, has been detected using an optical biosensor (Eyvazi et al., 2021).

Table 2
Comparison among various spectroscopic techniques.

Technique	Principle	Typical Information Obtained	Sample Preparation	Advantages	Limitations	Representative Applications
UV–Vis Spectroscopy	Measures the absorbance of amino acids or derivatives at specific UV/visible wavelengths.	Quantitative detection of derivatized amino acids; concentration estimation.	Hydrolysis required, derivatization (ninhydrin/OPA), dilution	Simple, cost-effective, suitable for routine analysis.	Requires derivatization; limited specificity in complex matrices.	Determination of total/free amino acids in hydrolyzed protein samples.
Fluorescence Spectroscopy	Detects emission from fluorescent amino acids or labeled derivatives.	Sensitive detection of aromatic amino acids (Tryptophan, Tyrosine, Phenylalanine).	Hydrolysis or extraction, derivatization (if needed), filtration	High sensitivity; suitable for trace analysis.	Limited to fluorescent species; matrix interference possible.	Monitoring protein oxidation and conformational changes.
NIR and MIR (Vibrational) Spectroscopy	Measures molecular vibrations corresponding to functional groups.	Provides molecular fingerprint and secondary structure information.	None or minimal, drying / grinding, direct measurement	Rapid, non-destructive, minimal sample preparation.	Overlapping bands require chemometric analysis.	Quality control and compositional profiling of food proteins.
Raman (Vibrational) Spectroscopy	Detects inelastic scattering of monochromatic light.	Structural and conformational information of amino acids and peptides.	None or minimal, drying / grinding, hydrolysis or extraction, direct measurement	Non-destructive; compatible with aqueous samples.	Weak signal intensity; fluorescence interference.	Studying protein folding, aggregation, and processing effects.
NMR Spectroscopy	Explores the magnetic resonance of atomic nuclei in amino acid environments.	Structural elucidation and quantitative analysis.	Extraction (aqueous/organic), solvent exchange (D_2O), pH adjustment	Highly specific, provides molecular-level insight.	Expensive, time-consuming, requires large sample amounts.	Characterization of amino acid composition and metabolite profiling.
Mass Spectroscopy	Measures mass-to-charge ratios of ionized molecules.	Accurate molecular mass and structure determination.	Hydrolysis or extraction, cleanup (solid phase extraction), derivatization (optional)	High sensitivity, specificity, and multiplexing capability.	Requires skilled operation, extensive sample prep.	Identification and quantification of free and bound amino acids in foods.

OPA- O-Phthalaldehyde

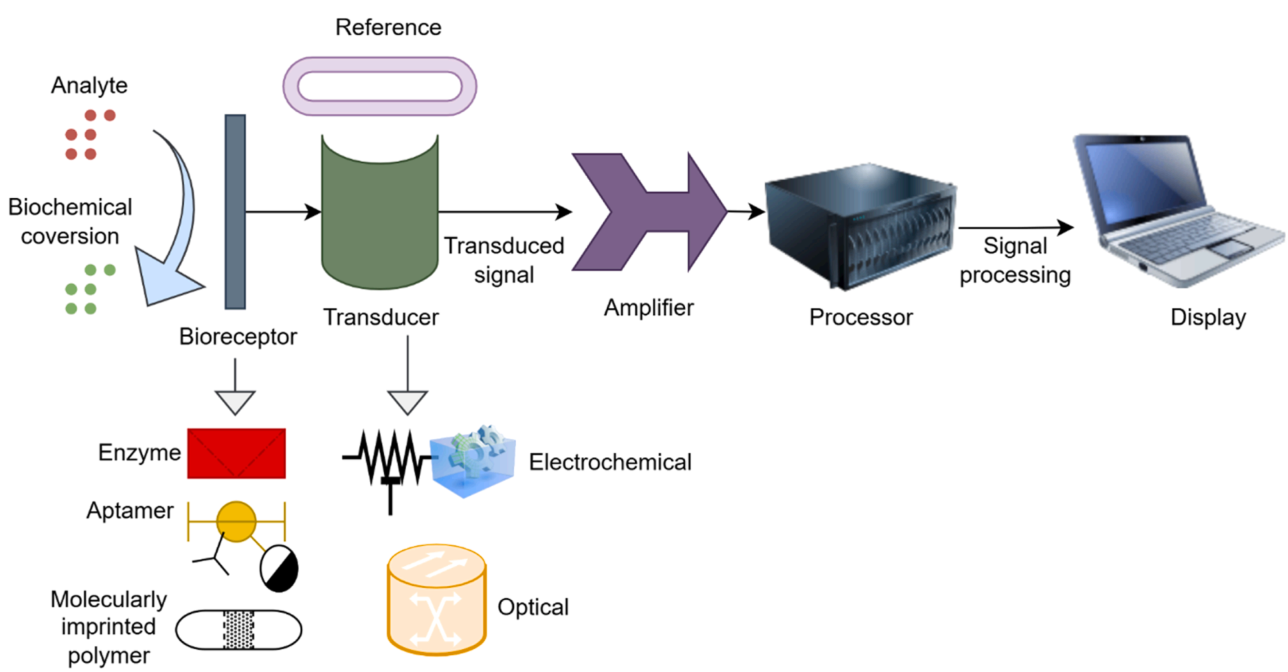

Fig. 6. Schematic diagram of a typical biosensor.

Although biosensing technologies have advanced rapidly, their application in amino acid detection within food systems remains limited. Most existing biosensors have been developed for monitoring food contaminants such as pesticides, toxins, pathogens, and biogenic amines, where the fundamental principles of bio-recognition and signal transduction are well established. These platforms commonly employ enzyme-, aptamer-, or molecularly imprinted polymer (MIP)-based recognition elements coupled with electrochemical or optical transducers, approaches that can be effectively adapted for amino acid analysis (Bunney et al., 2017; Eyvazi et al., 2021; Malik et al., 2023a; Turasan et al., 2021). The following sections classify biosensors according to 1) the key biorecognition elements and 2) transduction mechanisms for amino acid detection, followed by a discussion of emerging applications in food protein analysis.

4.2. Biorecognition elements

The biorecognition element, also known as a bioreceptor, is the key component of a biosensor that ensures selective interaction with the target analyte. It provides the specificity required for accurate detection through strong and selective binding. Various biorecognition elements, including enzymes, antibodies, aptamers, and molecularly imprinted polymers, exhibit distinct characteristics that influence biosensor performance. Understanding these properties is essential for designing reliable and sensitive detection systems (Morales & Halpern, 2018; Ravina et al., 2022). However, only a limited number of biorecognition elements have demonstrated practical applicability for detecting amino acids in food systems. The following section examines the most suitable options based on their specificity, stability, and compatibility with complex food matrices.

4.2.1. Enzyme-based recognition

Enzyme-based biosensors utilize specific enzymes as biorecognition elements to catalyze reactions with target amino acids. Common examples include amino acid oxidases and dehydrogenases, which selectively interact with residues such as L-glutamate, L-threonine, L-proline, and L-tyrosine (García-Guzmán et al., 2018; Pu et al., 2025; Tang et al., 2022). The enzymatic reaction produces measurable products, which can be detected electrochemically or optically. Advantages include high specificity and fast response times, although enzyme stability can be affected by pH, temperature, and matrix components. Immobilization strategies on electrodes or membranes can improve reusability and compatibility with complex food extracts (Cheng et al., 2023).

4.2.2. Aptamer-based recognition

Aptamer-based biosensors utilize short single-stranded DNA or RNA sequences that fold into specific three-dimensional structures, allowing for selective binding to target amino acids. Binding induces conformational changes that are transduced into detectable optical or electrochemical signals, such as photoinduced electron transfer, fluorescence resonance energy transfer, or excited state intramolecular proton transfer. They offer advantages over enzymes, including high stability (remaining effective in a wide range of conditions, including various pH levels and salt concentrations), ease of synthesis, and lower cost. Challenges include interference from complex food matrices, nonspecific binding, the need for careful aptamer selection, and the need to optimize for consistent performance under variable food-processing conditions (Wang et al., 2022a, 2022b; Yang et al., 2025). Aptamer-based recognition has been explored for only a limited set of targets, namely L-tryptophan, L-histidine, L-arginine, L-phenylalanine, and L-lysine, predominantly using peptide aptamer sensors (McKeague & Derosa, 2012; Song et al., 2008; W. Wang et al., 2023).

4.2.3. Molecularly Imprinted Polymer (MIP)-based recognition

MIP-based biosensors rely on synthetic polymers engineered to contain binding sites complementary in shape and functional groups to the target amino acid. During polymerization, the amino acid acts as a template, leaving cavities after removal that selectively rebind the target. MIPs can be integrated with electrochemical or optical transducers to convert binding events into measurable signals. MIP-based offers high stability, reusability, and resistance to harsh processing conditions. Limitations include slower binding kinetics compared to biological receptors and potential nonspecific interactions, which can be minimized through careful polymer design and surface functionalization (Akgönüllü et al., 2023; Ayerdurai et al., 2023). MIP-based biosensors are generally constrained to single-target recognition per polymer, including tryptophan, glycine, alanine, phenylalanine, and L- cysteine, based on template-driven detection methods (Chen et al., 2016; Gao et al., 2026; Haupt & Mosbach, 2000).

4.2.4. Comparison of biorecognition elements

Table 3 summarizes the key biorecognition elements used in biosensors for detecting amino acids in food matrices, highlighting their specificity, stability, and regeneration capabilities. Enzyme-based biosensors, using enzymes such as amino acid oxidases and dehydrogenases, offer high selectivity and rapid responses to amino acids, including glutamate, phenylalanine, and tyrosine. However, their performance can be influenced by factors like pH, temperature, and matrix components. Aptamer-based sensors provide chemical stability, ease of synthesis, and reversible binding, enabling multiplexed detection in complex food extracts. MIPs serve as robust synthetic alternatives, offering high thermal and chemical stability, which makes them suitable for harsh processing environments and reusable sensor platforms. The selection of a suitable biorecognition element depends on the target amino acid, required sensitivity, and the complexity of the food matrix. While enzymes provide superior specificity, aptamers and MIPs offer enhanced stability and versatility.

4.3. Classification based on transduction mechanism

Biosensor platforms for the detection of amino acids in food systems can be broadly categorized into two groups, electrochemical and optical biosensors, based on their transduction mechanisms, each offering distinct advantages and design challenges.

4.3.1. Electrochemical sensors (amperometric, potentiometric, conductometric)

Electrochemical biosensors are among the most suitable for detecting amino acids due to their inherent sensitivity, relatively low cost, and compatibility with miniaturized electronics. They convert biochemical interactions, such as those between enzymes, aptamers, or MIPs and their target substrates, into measurable electrical signals by detecting changes in charge distribution. Electrochemical biosensors typically employ a three-electrode system comprising a reference electrode (typically Ag/AgCl) to maintain a stable potential, a working electrode for signal generation, and a counter electrode to complete the circuit. In an amperometric sensor, the current resulting from an enzymatic or redox reaction is monitored (typically at a fixed potential). Potentiometric sensors measure changes in electric potential at the sensor surface with respect to the reference electrode, and conductometric sensors detect changes in conductivity (Eyvazi et al., 2021). Enzyme- and polymer-based electrochemical biosensors have been developed for detecting amino acids such as phenylalanine, tyrosine, and tryptophan in food and beverage samples, offering high selectivity and rapid response times (Cheng et al., 2023; Dinu & Apetrei, 2022).

Design considerations for electrochemical sensors for amino acids

Table 3
Comparison of biorecognition elements.

Bioreceptor type	Specificity	Stability (food-matrix)	Regeneration	Sample Preparation	Amino acids detected
Enzyme-based	Very high (substrate-specific)	Moderate (sensitive to inhibitors, pH)	Often limited (one-time use or limited cycles)	Buffer extract, filter/centrifuge, enzyme reagent addition	L-glutamate, L-threonine, L-proline, and L-tyrosine
Aptamer-based	High, modifiable	Good (thermal stability)	Good (can be denatured/ regenerated)	Buffer extract, sample clarification, aptamer incubation	L-tryptophan, L-histidine, L-arginine, L-phenylalanine, and L-lysine
MIP-based	Moderate to high	Excellent (synthetic)	Good (template removal)	Solvent extraction, sample clarification and finally load on MIP	Tryptophan, Glycine, alanine, phenylalanine, L-cystine.

MIP - Molecularly Imprinted Polymer

include immobilization of the bioreceptor (enzyme, aptamer, MIP), minimization of interfering species, optimization of the working potential (to reduce background current and interference from food matrix components), and long-term stability for repeated measurements (Eivazzadeh-Keihan et al., 2018). Robust sensors must also handle the complex food extract matrix. The combination of nanomaterials (e.g., carbon nanotubes, metal nanoparticles) with enzyme immobilization has enhanced sensitivity and lowered detection limits. For instance, in the detection of D-amino acids, a biosensor using D-amino acid oxidase immobilized on multi-walled carbon nanotubes coated with copper nanoparticles achieved very low detection limits (0.001 mM) and long-term stability (used 120 times in 2 months when stored at 4 °C) (Lata et al., 2012).

4.3.2. Optical biosensors (fluorescence, SPR, colorimetric)

Optical biosensors rely on changes in light absorption, fluorescence, scattering, or resonance changes induced by the analyte-bioreceptor interaction. Fluorescence biosensors employ fluorescent labels or detect changes in the fluorescent signal when amino acids bind to aptamers or enzymes (Yousefi et al., 2019). SPR sensors detect changes in refractive index at a metal-film interface, enabling label-free detection (Liu et al., 2020a, 2020b). Colorimetric sensors provide a visual signal change, often via enzyme-mediated color change or nanoparticle aggregation, for simplicity.

Optical biosensors can detect amino acid derivatives formed during protein degradation or processing, such as biogenic amines, oxidized sulphur compounds, and cross-linked residues, which serve as indicators of spoilage or adulteration. Aptamer-based fluorescence sensors utilize target-induced conformational changes to generate measurable optical signals, whereas gold nanoparticle colorimetric sensors rely on plasmonic shifts triggered by amino acid binding or nanoparticle aggregation. These systems enable rapid, label-free, and in situ monitoring of food quality and authenticity. Challenges remain in matrix interference (turbidity, colour, scattering), miniaturization, and achieving low detection limits in complex food extracts (Ha Anh et al., 2022; Huang et al., 2018; Lee et al., 2018; Xiong et al., 2021; F. X. Zhang et al., 2002).

4.4. Sample preparation and data analysis

Sample preparation is a critical step in biosensor-based amino acid detection, particularly when analyzing complex food matrices. Biosensors are typically employed for the quantitative determination of free amino acids, whereas protein-bound amino acids require prior hydrolysis to release the constituent residues before they can be detected. Commonly used pretreatment methods include homogenization, filtration, and centrifugation to remove particulates and reduce matrix interference (Alvarez Cañarte et al., 2025; Lee et al., 2018; Torre et al., 2019). For accurate quantification, enzymatic hydrolysis or acid digestion may be employed to release free amino acids from proteins and peptides. Selective extraction techniques, such as solid-phase extraction and polymer-based extraction, can further isolate target amino acids, thereby enhancing sensor sensitivity and specificity (Ashley et al., 2017). Data analysis involves converting the biosensor's raw signal, such as current, voltage, impedance, absorbance, or fluorescence, into meaningful quantitative information. This is typically achieved by constructing calibration curves using known concentrations of target amino acids. Advanced signal processing techniques, including baseline correction, noise filtering, and multivariate statistical analysis, are often employed to improve accuracy and mitigate matrix effects (Achmad et al., 2023; Dinu & Apetrei, 2022; Kaur et al., 2023).

4.5. Applications

While biosensor applications for amino acid detection and quantification in food remain limited, a substantial body of research exists for detecting food contaminants, monitoring food safety, and analyzing food adulterants. The principles, design strategies, and transduction mechanisms demonstrated in these studies highlight the considerable potential of biosensors for amino acid analysis. Building on these approaches, biosensor platforms can be adapted for food protein applications, enabling the detection and quantification of amino acids to assess protein quality. This section presents selected studies that illustrate how existing biosensor technologies can be leveraged to meet the growing need for rapid, sensitive, and in situ detection of amino acids in complex food matrices.

4.5.1. Cereals

Alvarez Cañarte et al., (2025) developed a biosensor using lysine alpha oxidase as the biorecognition element and a potentiometric transducer to detect lysine in meat sausages enriched with cereals and legumes. The sensor measured oxygen consumption during lysine oxidation, exhibiting high affinity ($K_m = 0.32$ mM), excellent sensitivity (limit of detection = 0.01 mM), and stability for 70 days, with the immobilized enzyme being reusable up to 18 times. Lysine quantification correlated strongly with HPLC results, demonstrating minimal environmental impact and cost-efficient operation. Tertis et al., (2023) developed a nanostructured graphite-gold platform to construct an electrochemical aptasensor for gliadin, the gluten component responsible for celiac disease. The aptasensor employed a screen-printed carbon working electrode functionalized with gold nanoparticles for covalent immobilization of a 40-base DNA aptamer, and indirect detection was based on a ferrocene electrochemical signal. The sensor exhibited high sensitivity and selectivity, with a detection range of 10–50,000 ng/L and a limit of detection of 3.4 ng/L, allowing accurate gluten detection in wheat flour and gluten-free products. Both studies demonstrate that enzyme- and aptamer-based electrochemical sensors can provide rapid, accurate, and environmentally friendly alternatives to traditional methods for monitoring key nutritional and allergenic components in foods.

4.5.2. Dairy

Researchers developed enzymatic biosensors for the quantification of lysine in dairy products, aiming to provide a faster, cost-effective, and environmentally friendly alternative to conventional HPLC methods. Villavicencio Yanos et al., (2025) developed a sensor that used enzymes immobilized on nylon and yucca bipolymer membranes as the biorecognition element, combined with an amperometric transducer to measure lysine in casein hydrolysates. This system demonstrated high affinity ($K_M = 0.37$ mM), enzymatic stability (trypsin remained active for approximately 35 days and could be reused up to 10 times), and a low environmental impact, allowing for rapid and accurate lysine quantification comparable to that achieved with HPLC. (Jadán Piedra et al., 2023) employed lysine-α-oxidase as the biorecognition element in a soluble enzyme sensor paired with an oxygen electrode as the transducer to detect lysine in mozzarella cheese. The sensor monitored oxygen consumption during lysine oxidation, achieving a rapid response time of 3.5 s and excellent specificity ($K_M = 0.03$ mM) over eight weeks of cheese ripening. Both studies successfully demonstrated that enzymatic biosensors could provide reliable, selective, and efficient measurements of lysine in dairy products, offering practical alternatives to standard HPLC analysis.

4.5.3. Fruits

Tang et al., (2022) developed an enzymatic glutamate sensor aimed at in vivo and on-site plant monitoring, integrating glutamate oxidase on a nanocomposite-modified screen-printed electrode composed of Au–Pt nanoparticles, carboxylated graphene oxide, and carboxylated multi-walled carbon nanotubes. This Nafion/GlutaOx/GO–COOH–MWNT–COOH/Au–Pt/SPE sensor demonstrated the widest detection range reported to date (2 μM–16 mM) and a low detection limit of 0.14 μM, enabling accurate in vivo glutamate determination in tomatoes across varying pH levels. Tasić et al.,(2022) developed a low-cost, eco-friendly electrochemical sensor for L-tryptophan detection by reusing graphite rods from discarded zinc-carbon batteries. Using cyclic and differential pulse voltammetry, the unmodified graphite electrode exhibited an irreversible two-electron, two-proton oxidation process with a detection limit of 1.73 μM, demonstrating excellent applicability in milk and apple juice samples. Mokole et al., (2024) developed a tryptophan sensor based on glassy carbon electrodes coated with CuO–polyaniline nanocomposites (made by both green and chemical synthesis). This sensor exhibited high conductivity and sensitivity, could detect tryptophan at concentrations as low as 2.037 μM, and performed reliably in pineapple samples (85–109% recovery). Collectively, these studies demonstrate the potential of nanostructured, enzymatic, and sustainable materials for developing reliable, cost-effective biosensors for agricultural and nutritional monitoring.

4.5.4. Fish

Researchers have developed rapid, sensitive, and cost-effective sensors for detecting histamine in fish, an important indicator of freshness and potential allergenic risk. Lee et al., (2018) fabricated a non-enzymatic electrochemical sensor using flake-like $Cu_3(PO_4)_2$ nanostructures formed on copper-electrodeposited screen-printed carbon electrodes and coated with 1% Nafion. This sensor exhibited good selectivity due to the negatively charged Nafion layer, a linear detection range of 5–500 ppm, a detection limit of 3 ppm, and accurate results for 45 h-rotten fish samples compared to LC-MS measurements. Torre et al., (2019) developed a simple enzymatic sensor by employing diamine oxidase immobilized on screen-printed carbon electrodes via glutaraldehyde and bovine serum albumin cross-linking. Using chronoamperometry, the sensor achieved a rapid response (60 s), a linear range of 1–75 mg/L, excellent reproducibility (RSD = 2.6%), and successful histamine detection in fish extracts. Both approaches highlight the potential of screen-printed carbon electrode-based sensors for fast, low-sample-volume, environmentally friendly, and on-site analysis of histamine in the food industry.

4.6. Emerging trends and knowledge gap

Recent developments in micro- and nano-technologies are rapidly transforming the biosensor field, driving trends toward miniaturization, higher sensitivity, multiplex detection, and seamless integration with smart, portable devices. Nanostructured materials such as carbon nanotubes, graphene, quantum dots, and metal (Au, Ag, Cu) or metal-oxide based nanoparticles have become central to enhancing sensor performance due to their high surface area, excellent conductivity, enhanced electron transfer kinetics, and novel binding modalities (Malik et al., 2023a; Wang, 2012). In amino acid sensing, immobilizing enzymes or aptamers on carbon nanotube-metal nanoparticle composites has significantly improved detection limits and stability. For example, multi-walled carbon nanotubes functionalized with 3,4,9,10-perylene tetracarboxylic acid enabled high sensitivity and selectivity for the detection of D- and L-alanine in porcine kidney, achieving a detection limit as low as 1.0×10^{-8} M (Xia et al., 2016). Such advancements enable biosensors to detect even trace amounts of amino acids or oxidative markers in complex food matrices, a task that is difficult with conventional methods.

In parallel, paper-based biosensors have gained traction as low-cost, flexible, and disposable platforms ideal for food monitoring. Using reversal nanoimprint lithography, paper-based plasmonic sensors embedded with Ag, Au, or Au-Ag alloy nanoparticles achieved high transfer efficiency (~85%) and strong resonance wavelength shifts ($\Delta\lambda$ = 33 nm for putrescine, 24 nm for spermidine) for biogenic amine detection (Tseng et al., 2017). Similarly, dual-colorimetric paper sensors using bromocresol purple and bromothymol blue enabled rapid, non-invasive detection of spoilage gases in chicken meat samples via smartphone-based RGB analysis (Abo Dena et al., 2021). These portable platforms can be adapted for amino acid detection by integrating enzyme or aptamer layers that convert amino acid concentration into measurable optical changes, offering opportunities for smart packaging and consumer-level food quality monitoring.

Microfluidic integration is another powerful trend. Lab-on-a-chip devices allow precise handling of small sample volumes, automated mixing, and multiplexed detection, reducing analysis time and reagent consumption (Logesh et al., 2021). In a recent study, a microfluidic smartphone-based biosensor enabled one-step, wash-free detection of four bacterial ssDNA targets using preloaded graphene oxide and fluorescent probes, achieving detection within five minutes and high sensitivity (Xing et al., 2023). The same microfluidic design principles - preloading reagents, controlled flow, and rapid reaction kinetics - can be applied to amino acid detection, enabling multi-analyte sensing directly from hydrolyzed food extracts.

Smartphone integration continues to expand biosensor capabilities through onboard imaging, automated analysis, and wireless data sharing. Custom applications for colorimetric or chemiluminescent sensors already allow rapid, field-ready food evaluation (Abo Dena et al., 2021; Montali et al., 2020). However, challenges remain, including complex sample pretreatment, inconsistent matrix effects, and incomplete app-level data processing. Addressing these gaps, along with enhanced cloud connectivity, AI-driven analysis, and IoT integration, will enable next-generation portable biosensors capable of reliable amino acid detection across diverse food systems.

5. Comparison of chromatography, spectroscopy, and biosensor techniques for amino acid detection in foods

Among the analytical platforms available, chromatography, spectroscopy, and biosensor technologies are three primary techniques, each offering distinct advantages and limitations in terms of sensitivity, speed, and applicability. Their comparative evaluation is crucial for selecting suitable analytical tools for both research and industrial applications. Table 4 compares chromatography, spectroscopy, and biosensor techniques.

Understanding the relative strengths and limitations of various analytical techniques is crucial for selecting an appropriate method that suits specific research or industrial objectives. For instance, chromatographic techniques such as HPLC and GC remain the benchmark techniques for amino acid determination. They provide exceptional resolution, quantification accuracy, and reproducibility, making them highly reliable for complex food matrices. The strength of chromatographic analysis lies in its ability to separate and detect individual amino acids in mixtures with high accuracy and specificity. However, these methods are often constrained by labour-intensive sample preparation, derivatization requirements, and long analysis times. Additionally, the high cost of instrumentation and the need for skilled operators limit their feasibility for routine or on-site applications. Despite these challenges, chromatography remains the reference standard for validating emerging analytical technologies.

Spectroscopic techniques, including NIR, Raman, and MIR spectroscopy, have gained attention as rapid, non-destructive alternatives. They offer the advantage of minimal sample preparation and can provide qualitative or semi-quantitative insights into amino acid profiles and protein-related changes in foods. These methods are valuable for high-throughput screening and quality control due to their simplicity and speed. However, their sensitivity is often lower compared to chromatography, and data interpretation can be complicated by overlapping signals and matrix interferences. Advanced chemometric modelling is typically required to extract meaningful information, which adds a computational layer to the analytical process. Consequently, spectroscopy is best suited for preliminary screening or process monitoring rather than precise quantification.

In contrast, biosensor technologies represent a new generation of analytical tools combining biological recognition with signal transduction. These devices convert specific biochemical interactions into

Table 4
Comparison between chromatography, spectroscopy, and biosensor technologies for amino acid detection and determination.

Parameter	Chromatography	Spectroscopy	Biosensor
Principle	Separation based on polarity/charge with detector quantification	Interaction of electromagnetic wave with molecular vibrations	Biorecognition element converts analyte interaction to signal
Sample preparation	Extensive (hydrolysis, derivatization)	Minimal to moderate	Minimal to moderate
Detection time	30–120 min	Seconds to minutes	Seconds to minutes
Sensitivity	Very high (ng/mL)	Moderate	High (μM-nM range)
Selectivity	Excellent	Moderate (needs chemometrics)	High (based on biorecognition element)
Instrumentation cost	High	Moderate to high	Low to moderate
Portability	Poor	Moderate	Excellent
Matrix interference	Low (after extraction)	High	Moderate to high
Suitability for on-site detection	Low	Moderate	High
Typical applications	Amino acid profiling, nutritional analysis	Protein structure, quality monitoring	Rapid screening, food authentication, spoilage detection
Amino Acids identified	Amino acids including cystine, alanine, valine, leucine, isoleucine, serine, threonine, glycine, methionine, aspartate, glutamate, lysine, arginine, histidine, phenylalanine, tyrosine, tryptophan, and proline	Limited set, usually aromatic amino acids (tryptophan, tyrosine, phenylalanine) and total amino acid content (UV/fluorescence/IR-based)	Highly selective and limited based on the bioreceptor type: L-glutamate, L-lysine, L-arginine, L-histidine, tryptophan, phenylalanine, glycine, alanine, L-cystine

measurable electrical or optical outputs. Depending on the configuration, biosensors can achieve high specificity through the use of selective biorecognition elements such as enzymes, aptamers, or MIPs. Electrochemical and optical biosensors are particularly prominent, offering rapid detection, portability, and compatibility with miniaturized and automated systems. Their simplicity, low reagent consumption, and potential for real-time monitoring make them highly attractive for food analysis. However, biosensors often face challenges related to long-term stability, reproducibility, and potential interferences from complex food matrices. Continuous advancements in nanomaterials, microfluidics, and signal amplification strategies are addressing these limitations, improving sensor robustness and detection accuracy.

This review provides a comprehensive evaluation of amino acids across various techniques, compared with the existing literature. Table 5 summarizes the key relevant articles on amino acid detection in food proteins. These reviews focus on specific topics such as D-amino acids (Marcone et al., 2019), organism-based metabolic studies (Kaur et al., 2023; Xu et al., 2020), or food-fraud detection (Hong et al., 2017), demonstrating the depth of traditional instrumental approaches but maintaining a strong laboratory-based approach. Collectively, they underscore the need for robust, sensitive, and reliable analytical workflows capable of handling diverse biological and food matrices. In this context, our review broadens the perspective by integrating classical analytical techniques with the latest advances in chromatography, spectroscopy, and biosensors, with a focus on amino acid detection. It uniquely emphasizes future-ready, portable, and real-time amino acid detection strategies to bridge the gap between laboratory performance and practical food-industry applications.

6. Conclusion

In summary, chromatographic, spectroscopic, and biosensor-based platforms each contribute uniquely to the detection and determination of amino acids in foods. Chromatography remains the gold standard for quantitative and compositional analysis, while spectroscopy provides a rapid, non-destructive approach ideal for process control and high-throughput screening. Biosensors, although still developing for amino acid applications, offer potential for in-situ and real-time monitoring due to their portability, specificity, and adaptability to smart analytical systems. The choice of technique ultimately depends on analytical goals, sample complexity, and available resources. For selecting a rational method, a systematic comparison of key analytical parameters, such as sensitivity, selectivity, reproducibility, cost, and throughput, is

Table 5
Comparison of relevant review articles for the detection of amino acids in food proteins.

Study	Focus area	Techniques covered	Scope of application	Compared to this study
Modern analytical techniques for food fraud and adulteration (Hong et al., 2017)	Detection of adulteration in highly adulterated food categories	Chromatography, spectroscopic, electrophoretic, inductively coupled plasma and immunoassays methods	Adulteration detection, species/origin verification, food-fraud surveillance	Emphasis on traditional techniques; our study expands to portable biosensors, nanomaterials, and real-time detection of amino acids specifically
Analytical methods for amino acid determination in biological organisms for food and human research (Xu et al., 2020)	Broad survey of amino acid analysis methods across biological and food-related samples	Chromatography, capillary electrophoresis, NMR, amino-acid analyzers, electrochemical sensors	Nutritional analysis, broad sample types	Provides traditional methods for amino acid analysis; our review builds upon this by adding spectroscopic and emerging biosensors for practical, real-world applications, specifically for analysis of amino acids in foods
D-amino acids in foods (Marcone et al., 2019)	Occurrence and detection of D-amino acids in foods	Chromatographic separation, derivatization, and biosensors detection of D-amino acids	Nutritional safety analysis of food proteins	Narrow focus on D-amino acids; our review covers the detection of all amino acids, including spectroscopic techniques; advances toward biosensors and field-level detection
Recent analytical technique trends in amino acid detection in biological samples (Kaur et al., 2023b)	Analytical and bioanalytical methods trending in amino acid detection	Chromatography and electrochemical sensors	Food, pharmaceutical, biological samples	Comprehensive on established and some sensor-based methods; our review emphasizes spectroscopic and next-gen sensors, for amino acid detection
Current study	Inclusive overview from traditional to emerging techniques for amino acid detection in food proteins	Chromatography, spectroscopy, and biosensors	Food proteins across matrices; real-time, portable amino acid analysis	Broader scope; integrates established methods with emerging techniques; outlines gaps and future directions for rapid, high-throughput food-based amino acid analysis

essential.

Future research should prioritize the development of biosensor techniques tailored to plant-based foods, as rising consumer demand for plant-derived proteins necessitates rapid, on-site evaluation of protein quality during processing and storage. Additionally, comprehensive characterization of food samples will require an integrated analytical approach in which a single sample is analyzed using chromatography, spectroscopy, and biosensor-based methods to generate cross-validated, highly reliable amino acid profiles. Further advancements should also explore integrating microfluidic platforms with advanced spectroscopic and electrochemical detection to create fully automated systems capable of analyzing complex food matrices with minimal sample preparation. At the same time, building standardized spectral libraries and sensor-response databases for a wide range of food products will support the development of robust machine-learning prediction models, enhancing accuracy, instrument transferability, and broader adoption of these technologies in amino acid determination.

CRediT authorship contribution statement

Mohammad Nadimi: Writing – review & editing, Supervision. **Elham Salimi:** Writing – review & editing, Supervision. **Jitendra Paliwal:** Writing – review & editing, Supervision, Project administration, Funding acquisition. **Chitra Sivakumar:** Writing – original draft, Visualization, Methodology, Formal analysis, Conceptualization. **Sristi Mundhada:** Writing – original draft, Software, Methodology, Formal analysis, Conceptualization. **Anup Dey:** Writing – original draft, Conceptualization.

Declaration of Competing Interest

The authors declare that they have no competing financial interests or personal relationships that could have appeared to influence the work reported in the manuscript.

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